CLEANASAURUS · ALPHA

"STORAGE ALMOST FULL."

One developer ran a cleanup tool on their laptop and found **58.68 GB** in **128** old `node_modules` folders.

Most of them can be rebuilt with one command. Almost nobody deletes them, because deleting the **wrong** folder the night before a deadline costs you your project.

Source: Flavio Copes, ”npkill” (flaviocopes.com/npkill). Nate Kim · Nate Park · Wesley Chang

MVP VALIDATION · n = 10 (5 CS, 5 non-CS) · planned 15 · Oct 30 – Nov 2

Testers let it act. It didn't free the space they needed.

KPI, AS PITCHED	PASS	FAIL	RESULT	
Approve and run a **full** plan in session 1	≥ 60%	< 40%	**30%** (3/10) 70% ran at least 80% of a plan	FAIL
Run a second cleanup on their own	≥ 40%	< 20%	**30%** (3/10) measured over 3 days, not 7	ITERATE
Executed actions later undone	≤ 10%	> 25%	**15%** (41/267, one tester)	ITERATE
Unsafe actions reaching a user	0	any	**0**	PASS

Outcome: hypothesis not validated as built → PIVOT. Testers got back about **2%** of the space they wanted (0.6 of 27.5 GB, medians), and FREED showed 12.4 GB while **0 bytes** left the disk.

Source: our validation report (validation/MVP_VALIDATION.md), sessions run by Nate Kim. Thresholds are the ones in our MVP pitch.

PIVOT: CS STUDENTS' REBUILDABLE DEV CLUTTER

38 GB

median rebuildable dev clutter per CS tester

1.4 GB

median duplicates our MVP could free

5 of 5 CS testers said their real space was somewhere the MVP wouldn't go.

The competition, honestly

CleanMyMac CLI (MacPaw, developer preview) already purges `node_modules` older than 7 days and clears Xcode DerivedData. npkill and DevCleaner each do one ecosystem.

What none of them do

- **Prove** a folder is rebuildable: lockfile, gitignored, no uncommitted work, no secrets
- Save the exact rebuild command
- Let Claude map "I'm done with CS 3140" to projects

LIVE DEMO · END TO END · 3½ MIN

One request, from a full disk to space back

1

Open a messy `~/code`: 8 projects, a VM, one with a `.env` secret

2

Type: "I'm done with CS 3140 and the hackathon. Free up space."

3

Claude matches "CS 3140" and "hackathon" to projects and writes the plan

4

Validator re-checks every folder. Live, Claude left out the hackathon (uncommitted work) and asked. When Claude slips, the validator rejects it and Claude retries

5

Approve: space freed, measured on disk. Undo shows the rebuild commands

Claude decides

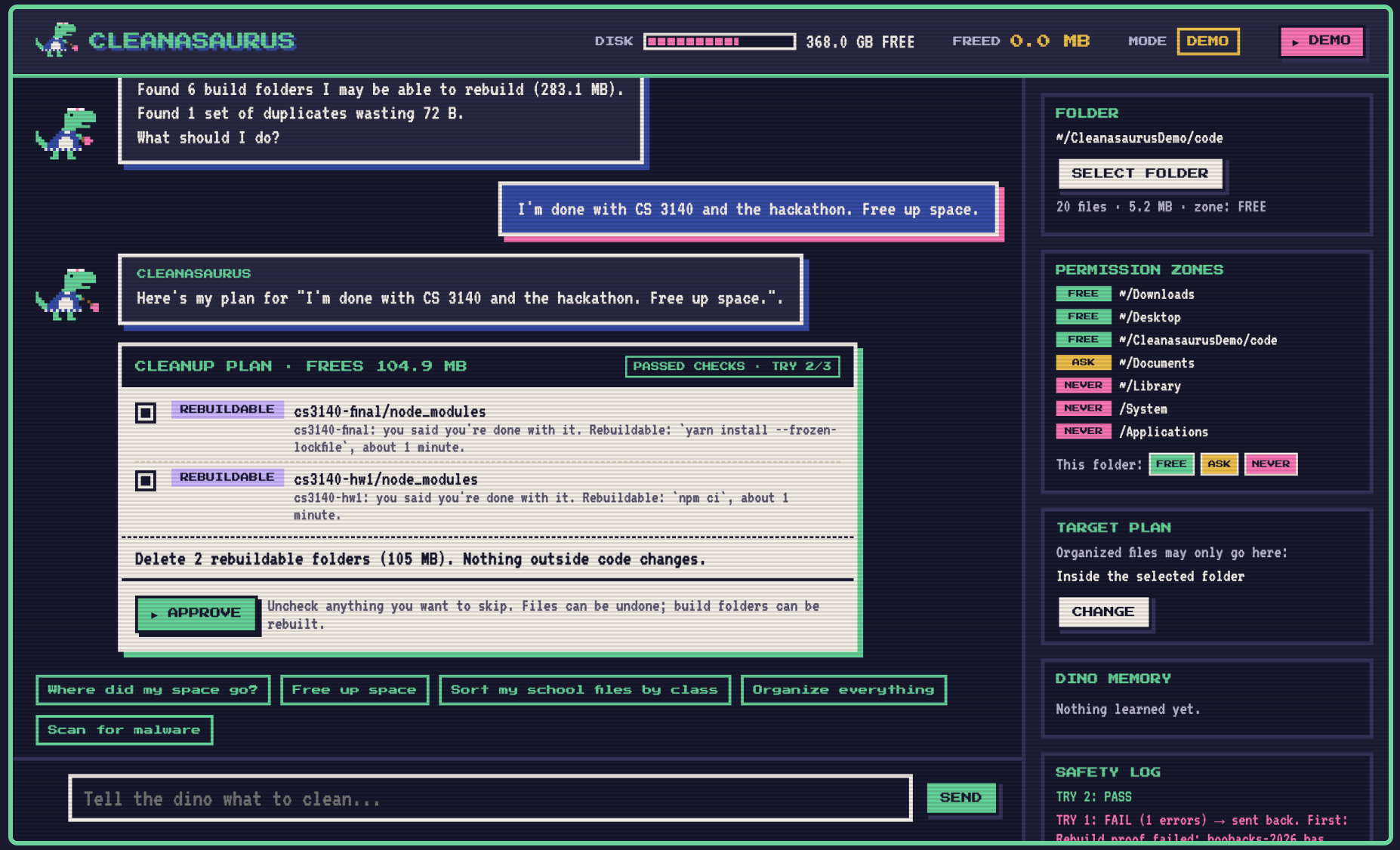
Which projects "done with CS 3140" means · how to explain each action · when to ask instead of guess

Plain code decides

Whether a folder can be deleted (the rebuild proof) · what was really freed · nothing runs until you approve

Tricky data on purpose: uncommitted work, a project edited 4 days ago, a `.env` inside `node_modules`, a lockfile-less project, a VM disk. Backup: the same request on the offline planner (next slide).

BACKUP IF THE LIVE DEMO FAILS



The scripted request, offline planner: only CS 3140 and hackathon projects are proposed → hoohacks-2026 rejected (uncommitted work) → try 2 passes with the two CS 3140 folders, each with its rebuild command. The other courses, the .env secret, and the VM are never touched.

METHODOLOGY & ENGINEERING METRICS

30
user stories

29 / 18
GitHub issues logged / closed
6 from the MVP, 12 in the Alpha · 1 partial · US-30 built, issue to file

141
automated tests, 105 written before their code
+ 22-step UI test · + 5 live tests on the real Claude: 5/5 pass

4.7
tests per story
3.8 counting only runnable tests · every story has ≥ 2

100%
runnable tests pass (113/113)
80% of all 141; 28 todo kept red, CI-checked

96.1%
backend line coverage
91.1% branch

25 revised
AI-generated tests added or changed after review, all logged · 3 bugs found by hand testing became 7 new tests · tests hand-edited so far: 0, team review before Beta

Tests first, no code (c1c337d) → Rebuild proof green (d2f885d, 454d384) → Claude-path + intent tests red → 113/113 green → Live Claude 5/5

Insight: tests, reviews, and hand testing caught 11 real bugs (REG-1…8 and 3 intent bugs) and 5 tests that couldn't fail. Every bug became a failing test before its fix.

STAGE & ROADMAP TO BETA

Alpha: done

✓ Validation run (n = 10), decision made: Pivot
✓ 30 stories with acceptance criteria → 29 GitHub issues (US-30 to file)
✓ Rebuild proof + JS and Python cleanup, real deletes
✓ Honest FREED (measured on disk)
✓ Clean by intent ("done with CS 3140 but keep the final", "not done with…"), caches, conda
✓ One-click rebuild, undo one action (and the folders a run made), Trash, one-line summary, storage breakdown (partial)
✓ Empty the Trash, only after a separate confirmation (US-30)
✓ Claude path: 7 simulated + 5 live tests on the real model (5/5); CI on every push

Beta: 4 weekly sprints, then a pivot test (dates proposed)

1. Nov 9: protect projects (US-04) · record the Claude demo · live tests each release
2. Nov 16: Xcode and Docker cleanup
3. Nov 23: ML caches · storage watch
4. Nov 30: Go Pro fake door · recruit 20 testers · start pivot test

BETA PIVOT TEST: 20 CS STUDENTS WE DON'T KNOW	TARGET
First-session approval of a plan freeing ≥ 10 GB	≥ 60%
Median real space freed	≥ 10 GB
One-click rebuilds that work · unrecoverable deletions	≥ 95% · 0
Moat: prefer us over CleanMyMac's purge, head to head	≥ 50%
"Go Pro" clicks / pre-orders	≥ 15% / 5%

QUESTIONS?

Claude writes the plan. Plain code proves it’s safe. You get the space back.

github.com/natekim416/cs4501_startup · tests: `cd app && npm test`